

Empirical analysis of machine learning classifiers for retinal and optic nerve disorder detection using band-limited PERG features

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ABSTRACT

Pattern Electroretinogram (PERG) signals offer critical insights into the functional status of retinal ganglion cells and are widely used for diagnosing optic nerve and retinal pathologies. While existing research has largely focused on binary classification or full-spectrum signal analysis, limited studies have addressed the use of frequency band-based feature extraction in multiclass classification. This study presents a novel methodology that segments PERG signals into defined frequency bands and extracts a robust set of 33 time-domain features per eye, encompassing statistical, morphological, and temporal characteristics. The dataset, originally comprising 15 diagnostic classes were consolidated into five clinically relevant pathology categories to enhance class balance and interpretability. Feature relevance was validated through one-way ANOVA, identifying key discriminative features (e.g., slope, std) with high F-values and $p < 0.001$. Several machine learning models were developed and evaluated, including Multilayer Neural Networks using the Levenberg–Marquardt algorithm (MLNN-LM), Random Forest, Bagged Trees, RUSBoost, and AdaBoost implemented via an Error-Correcting Output Codes (ECOC) framework. Among these, the AdaBoost-ECOC model demonstrated superior performance with an average classification accuracy of $99.11\% \pm 1.26$, precision of $82.92\% \pm 0.59$, and F1-score of $83.03\% \pm 0.55$. These findings highlight the clinical applicability of band-specific time-domain features, supported by statistical validation, in enabling accurate and efficient multiclass classification of PERG signals.

1. Introduction

Early detection of eye disorders is critical, yet clinical evaluation can be time-consuming and may delay diagnosis until symptoms become severe [1]. With the increasing availability of non-invasive diagnostics techniques, there is a strong interest in using artificial intelligence (AI) to support automated and objective eye disease detection [2]. Among these technologies, ocular electrophysiology offers a direct and quantitative assessment of retinal function by measuring electrical responses elicited by visual stimuli [3]. Core techniques like flash electroretinography (ERG) and visual evoked potentials (VEP) have long been used in clinical diagnosis. Flash ERG captures photoreceptor (a-wave) and bipolar cell (b-wave) activity using full-field stimuli, while VEP assesses

cortical responses and is useful for evaluating optic nerve integrity [4–8]. Although structural diagnostic tools such as optical coherence tomography (OCT) and visual field analysis are widely used, they often detect disease at later stages. In contrast, electrophysiological methods such as multifocal ERG (mfERG) and multifocal VEP can reveal functional impairments earlier in disease progression [9].

In ophthalmic applications, electroretinography (ERG) and its variant, pattern electroretinography (PERG), use corneal or skin electrodes to capture retinal responses to visual stimuli [10]. Pattern electroretinography (PERG) is particularly sensitive to early dysfunction of retinal ganglion cells (RGCs), making it clinically valuable in conditions such as glaucoma [11]. PERG employs contrast-reversing checkerboard stimuli focused on the macular region, producing characteristic P50 and

Abbreviations: AdaBoost, Adaptive Boosting; AI, Artificial Intelligence; AUC, Area Under the Curve; ECOC, Error-Correcting Output Codes; Electrocardiographic, (ECG); Electroencephalographic, (EEG); Empirical Wavelet Transform, (EWT); FFT, Fast Fourier Transform; FN, False Negative; FP, False Positive; Hz, Hertz; IOBA, Instituto Universitario de Oftalmobiología Aplicada; LM, Levenberg–Marquardt (training algorithm); MAV, Mean Absolute Value; ML, Machine Learning; MLNN, Multi Layer Neural Network; PERG, Pattern Electroretinogram; RF, Random Forest; RGC, Retinal ganglion cell; RMS, Root Mean Square; RUSBoost, Random Under Sampling Boosting; TN, True Negative; TP, True Positive.

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N95 components that reflect macular and inner retinal function, respectively [12,13]. Recordings are typically obtained following ISCEV-recommended protocols, though the signals remain susceptible to noise arising from eye movements, fixation instability, and electrophysiological interference [14–16].

Recent advances in signal processing and machine learning have significantly enhanced the analytical potential of PERG signals. Techniques such as multi-wavelet transformations and data-driven feature extraction have been shown to improve noise suppression and signal characterization in electrophysiological recordings [4,17]. For example, wavelet-based approaches using Ricker, Gaussian, and Morlet functions have achieved high classification accuracy by capturing complementary temporal–spectral characteristics of ERG signals [18]. These developments have facilitated improved detection of pathological patterns across a wide range of retinal and optic nerve disorders, including glaucoma, retinitis pigmentosa, and less common inflammatory and ischemic conditions [4].

Beyond ophthalmology, structured signal decomposition and feature-engineering frameworks have proven effective across multiple understanding biomedical domains. Empirical Wavelet Transform (EWT), Fourier–Bessel series expansion, and multivariate decomposition techniques have been successfully applied to electrocardiographic (ECG), electroencephalographic (EEG), and pulmonary sound signals for automated disease detection, including myocardial infarction, bundle branch block, epilepsy, and respiratory disorders [19–21]. These studies demonstrate that physiologically motivated band-wise decomposition, followed by interpretable feature extraction, can yield robust and clinically meaningful diagnostic models. Similar wavelet-based approaches have also been applied to ophthalmic imaging, such as automated glaucoma detection from fundus images, highlighting the broad applicability of multiscale signal analysis in biomedical diagnostics [22].

Pattern electroretinography (PERG) has consequently emerged as a critical tool for assessing macular and retinal ganglion cell function, particularly in distinguishing macular diseases from optic nerve disorders [23,24]. Recent advancements in signal processing and machine learning (ML) have expanded its diagnostic potential. For instance, Canedo et al. proposed novel modeling approaches for PERG signal analysis, leveraging wavelet transforms and time-frequency decomposition to enhance feature extraction for retinal pathology detection [25]. Similarly, studies such as Asanad and Karanjia emphasize PERG's clinical applicability in diagnosing conditions such as glaucoma and inherited maculopathies [23], while Holder et al. linked PERG abnormalities to fundus autofluorescence patterns in macular dystrophies [24]. The ISCEV standards (Bach et al., 2013) provide a standard procedure for consistent PERG acquisition, though gaps remain in translating these protocols into automated, multi-disease classification systems [15,26].

Despite these advances, most existing machine learning studies focus on binary classification tasks (e.g., glaucoma versus healthy controls) using ERG or OCT data, with relatively limited attention given to PERG-based multi-pathology classification [27]. Moreover, while wavelet-based and spectral features have demonstrated promise in ERG analysis, systematic evaluation of band-limited PERG signals for distinguishing diverse retinal and optic nerve disorders remains underexplored [28,29]. While prior studies have applied machine learning to diagnose individual retinal diseases, the majority rely on binary classification paradigms and limited datasets [30]. To address these limitations, this study utilizes the IOBA-PERG dataset that contains a comprehensive repository of 1354 transient pattern electroretinography (PERG) responses, to address broader diagnostic challenges through two key approaches [4]. Advanced signal processing techniques are applied to analyze waveform morphology, focusing on features such as N35–P50–N95 latencies, oscillatory potentials, and spectral characteristics. Drawing from established EEG signal processing methodologies—where frequency band segregation and band-pass filtering (0.5–100 Hz) have enabled robust feature extraction [31], this study

adopts a band limited PERG analysis framework to enhance interpretability. The PERG signals are segmented into five physiologically motivated frequency bands:

1. Very-low (0–1 Hz): Captures slow oscillatory components associated with metabolic retinal processes
2. Low (1–5 Hz): Isolates the N95 wave, indicative of inner retinal dysfunction
3. Mid (5–15 Hz): Emphasizes the P50 wave, representing macular function
4. High (15–30 Hz): Associated with transient glaucomatous responses
5. Very-high (30–50 Hz): May encode high-frequency oscillations in ischemic retinopathies

Unlike prior ERG and PERG studies that primarily apply spectral filtering as a preprocessing step or focus on isolated disease detection, the present work introduces a structured, band-wise diagnostic framework in which frequency-specific PERG components are independently analyzed, statistically validated, and systematically compared across multiple retinal and optic nerve disorders. The novelty therefore lies not in the use of standard band-pass filters, but in the diagnostic interpretation and comparative evaluation of band-limited PERG features across diverse pathologies.

Second, multi-class classification models are developed to distinguish among 15 retinal pathologies using time-domain features derived from band-limited PERG signals. The primary objective is to establish a machine learning-driven framework that incorporates frequency-optimized PERG features for the diagnosis of retinal disorders, including early-stage glaucoma, diabetic macular edema, and retinitis pigmentosa. By systematically evaluating the diagnostic relevance of predefined frequency bands, this study aims to address existing gaps in PERG-based classification and to highlight the advantages of frequency-adaptive filtering over conventional time-domain analysis.

Although existing literature confirms the clinical relevance of PERG signals for assessing retinal function, significant gaps remain in their analytical use for automated classification and early detection. In particular, frequency-domain analysis of PERG signals has received limited attention in machine learning-based ophthalmic diagnostics. This work addresses this gap by investigating the diagnostic value of PERG frequency components using an ML architecture trained on the open-access IOBA-PERG dataset from PhysioNet. The proposed framework, illustrated in Fig. 1, analyzes multi-pathology PERG responses, with implementation details and results presented in the subsequent sections.

2. Methodology

This section outlines the complete methodology adopted in this study, including signal preprocessing, band segmentation, feature extraction, and the feature validation. Each process is described in detail in the following subsections.

2.1. Dataset description

In this research, the publicly available IOBA-PERG dataset from PhysioNet [4] have been used, which comprises 1,354 transient pattern electroretinogram (PERG) responses recorded from 304 individuals, encompassing a total of 336 independent session records. Each record contains bilateral retinal signals sampled at a rate of 170 Hz, alongside extensive clinical metadata including age, sex, and multiple diagnostic labels (diagnosis1, diagnosis2, diagnosis3). In this analysis, only the primary diagnostic field (diagnosis1) was used to define the ground truth labels for supervised multi-class classification system.

Although the complete dataset includes a broad range of retinal and optic nerve pathologies, a curated subset consisting of 15 clinically significant disease categories, including healthy controls, have been

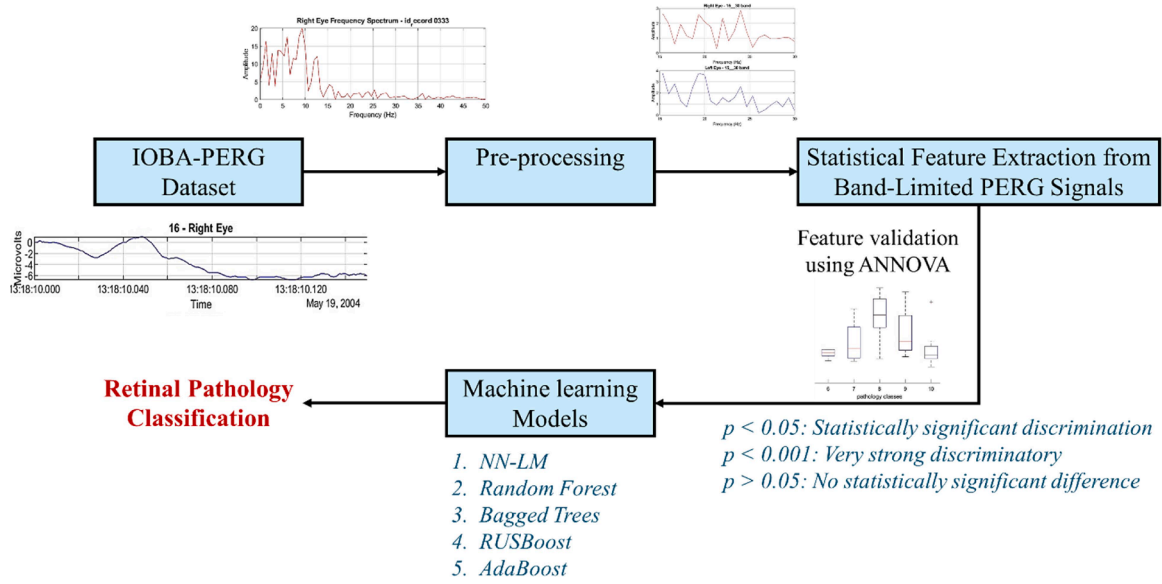


Fig. 1. Block diagram of the proposed classification of retinal pathologies using ML algorithms.

chosen. This selection was primarily motivated by the need to balance class representation and mitigate the adverse effects of sample size disparities. Categories with fewer than five recordings were excluded to prevent data imbalance, which can compromise classifier performance. Future work will aim to extend the analysis to include underrepresented classes once the present methodology has been validated.

The final dataset used for classification included 280 records, stratified across 15 diagnostic categories. Table 1 presents a summary of the most prevalent pathologies retained for analysis, along with brief clinical descriptions.

Table 1
Summary of selected diagnoses from the IOBA-PERG dataset.

Sl. No.	Name of the Disease	No. of Records	Explanation
1	Normal	106	No clinical signs of retinal dysfunction.
2	Retinitis Pigmentosa	47	Genetic retinal dystrophy affecting photoreceptor response.
3	Macular Dystrophy	33	Degenerative or inherited disorder of the macula.
4	Stargardt Disease	16	Juvenile macular degeneration caused by ABCA4 mutations.
5	Cone-Rod Dystrophy	14	Affects cone and rod photoreceptors, leading to progressive vision loss.
6	Birdshot	12	Rare bilateral posterior uveitis characterized by creamy lesions.
7	Chorioretinopathy	10	Genetically mediated degeneration of retinal ganglion cells.
8	Inherited Optic Atrophy	8	Non-progressive dysfunction of rod photoreceptors.
9	Congenital Stationary Night Blindness	7	Damage induced by systemic or intravitreal medications.
10–15	Retinal Toxicity	27	Autoimmune retinopathy, ischemia, achromatopsia, etc.
	Other Rare Disorders [†]		
	Total	280	

[†] Other Rare disorders: autoimmune retinopathy, orbital ischemia, X-linked retinoschisis, central serous chorioretinopathy, congenital achromatopsia, and paracentral acute middle maculopathy

2.2. Preprocessing of PERG signals

All PERG recordings were initially subjected to preprocessing to enhance signal quality and minimize noise. The signals, acquired at 170 Hz, capture the dynamic activity of retinal cells within the relevant electrophysiological frequency range (0.5–50 Hz). Each session provided both left and right eye recordings. The raw signals were depicted in Figs. 2 and 3, represents a sample normal PERG signal and Macular Dystrophy signal accordingly.

In addition, raw signals were filtered using a fourth-order zero-phase Butterworth bandpass filter (0.5–50 Hz) as shown in Eq. (1). This choice was guided by standard electrophysiological analysis practices [2], effectively suppressing baseline drift and high-frequency electrical artifacts without distorting phase information. The zero-phase implementation provide that temporal relationships between signal components are preserved [32]. This is applied separately to left and right eye signals.

$$x_{filt}(t) = \text{filtfilt}(b, a, x(t)) \quad (1)$$

where,

$$(b, a) = \text{Butterworth}(4, [f_L, f_H])$$

f_L and f_H are the normalized low and high cutoff frequencies

Post-filtering, spectral content was examined through Fast Fourier Transform (FFT) to assess signal quality and identify dominant frequency components [33]. This preprocessing step served both to denoise the data and to prepare it for subsequent band-specific decomposition and feature extraction. Fig. 4 presents a sample raw PERG signal, filtered and with FFT applied, the frequency components in the 0–50 Hz range.

2.3. Frequency band decomposition

While band-pass filtering and spectral analysis have been previously applied to ERG and PERG signals, most existing studies employ these techniques primarily for noise suppression or for single-disease characterization. In contrast, this study uses frequency band decomposition as a diagnostic analysis framework, in which each band is analyzed independently to examine its relative contribution to disease discrimination. This structured band-wise analysis enables systematic comparison of feature relevance across multiple retinal and optic nerve disorders, rather than treating spectral filtering solely as a preprocessing

PERG Signals - Normal

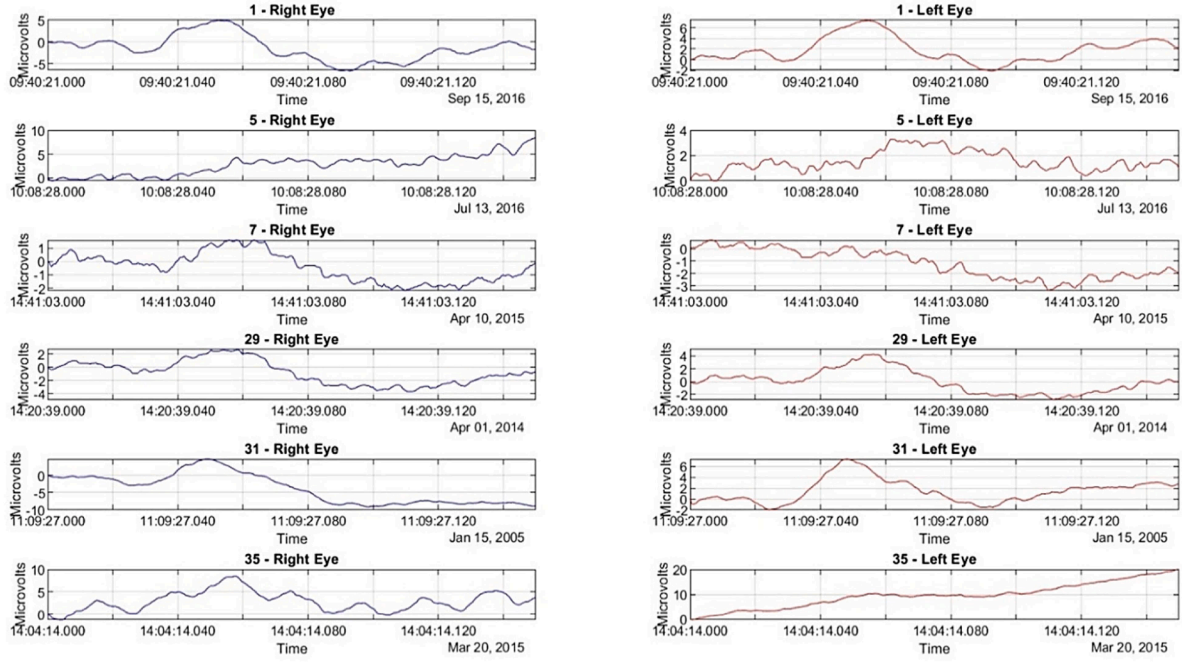


Fig. 2. Raw PERG signal (normal subject).

PERG Signals - Macular dystrophy

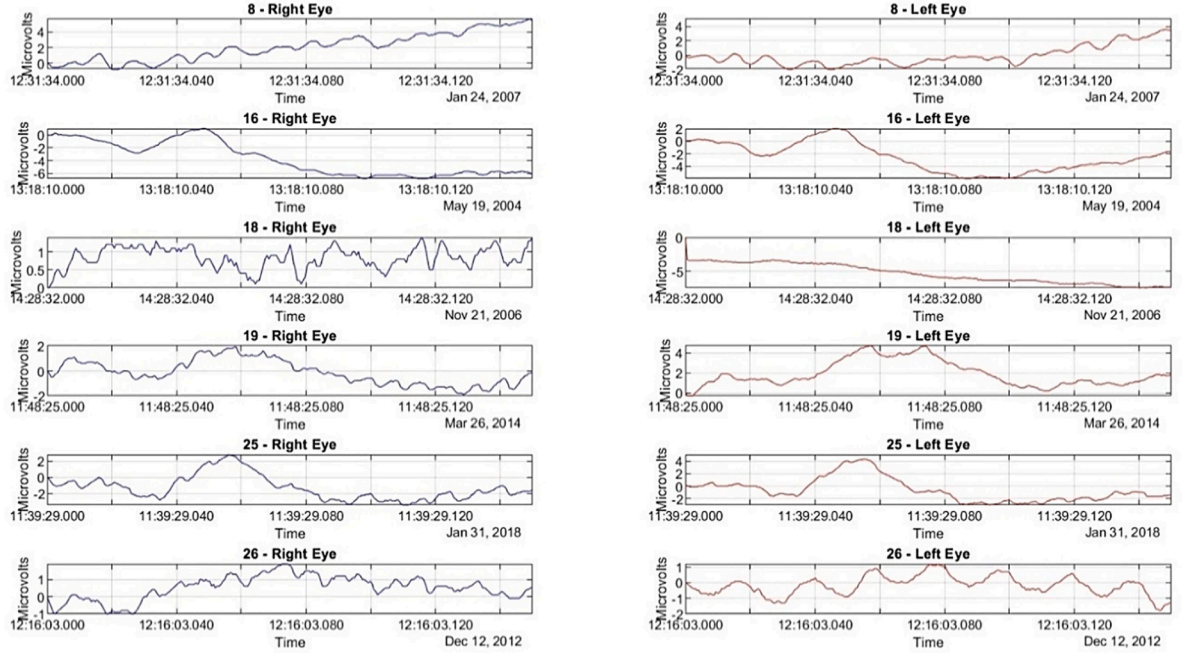


Fig. 3. Raw PERG signal (macular dystrophy).

step.

Subsequently, to isolate physiologically meaningful signal components, the filtered PERG signals were decomposed into five distinct frequency bands, each reflecting different retinal or neurophysiological phenomena [28]. Table 2 list the proposed frequency band limits along with their functional relevance. For each eye signal x_t , following bandpass filtering, the frequency spectrum and corresponding band segmentation were extracted as described in Eq. (2 and 3).

$$X(f) = |F\{x_{filt}(t)\}|, f = 0, 1, \dots, \frac{F_s}{2} \quad (2)$$

where,

- $X(f)$ is the magnitude of the Fast Fourier transform (FFT)
- $B_i = [f_{i, low}, f_{i, high}]$ is a defined frequency band (e.g., 1–5 Hz)

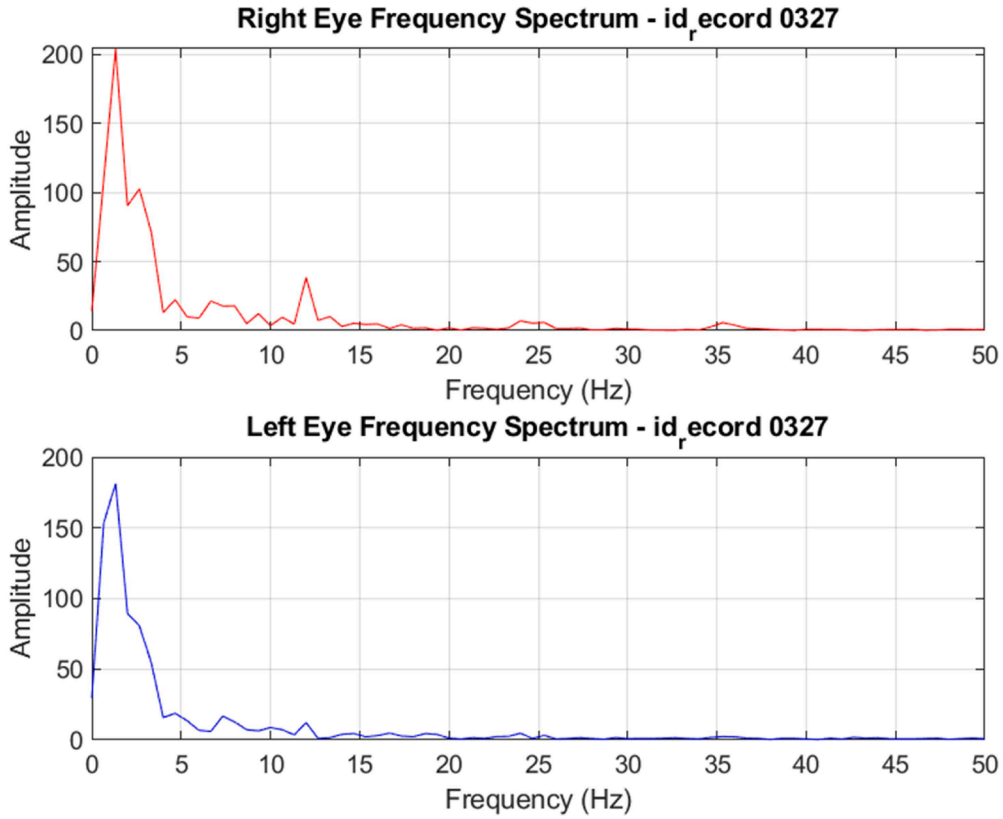


Fig. 4. Frequency Spectrum of a raw PERG signal (0–50 Hz).

Table 2

Frequency Band Range (Hz) - Functional Relevance.

Band		Frequency Range (Hz)		Interpretation
B_i	Band name	$f_{i,low}$	$f_{i,high}$	
1.	Very low	0	1	Baseline fluctuations, non-neural drift
2.	Low	1	5	Principal PERG components (e.g., N35, P50, N95)
3.	Mid	5	15	Flicker responses, ganglion cell activity
4.	High	15	30	Oscillatory potentials, secondary components
5.	Very high	30	50	Noise-prone region, possible residual oscillations

For each band B_i , extract:

$$X_i = \{X(f) | f \in B_i\} \quad (3)$$

Frequency spectra were analyzed for both eyes independently, and magnitude responses were retained for subsequent feature extraction. Figs. 5–9: depicts the Band-specific frequency plots for both eyes.

This band-based separation enables detailed feature analysis across different physiological domains.

2.4. Statistical feature extraction from band-limited PERG signals

Following frequency band decomposition, a set of statistical features was extracted independently from each selected frequency band and for each eye. Although these features are computed in the time domain, they are derived from band-limited PERG signals obtained through frequency-domain segmentation. This approach enables characterization of signal morphology and variability within physiologically meaningful frequency ranges while maintaining interpretability. For each frequency band and eye, a total of 11 scalar statistical features were

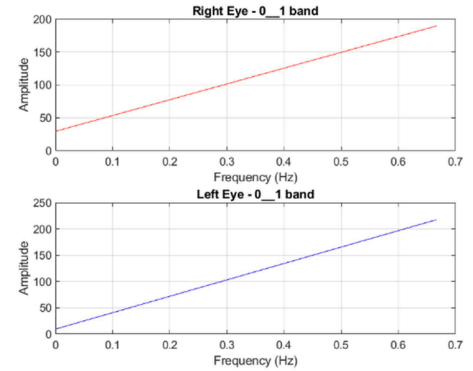


Fig. 5. Frequency plot - (0 to 1 Hz) Right and Left eye.

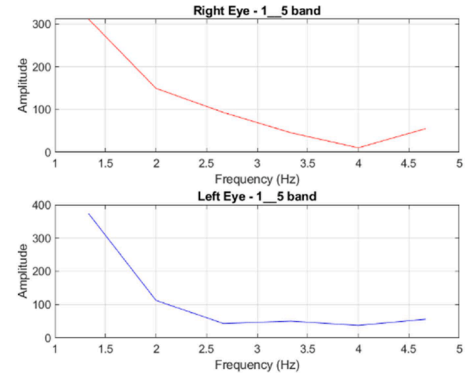


Fig. 6. Frequency plot - (1 to 5 Hz) right and left eye.

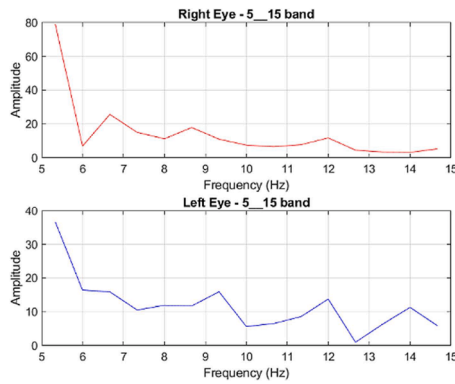


Fig. 7. Frequency plot - (5 to 15 Hz) right and left eye.

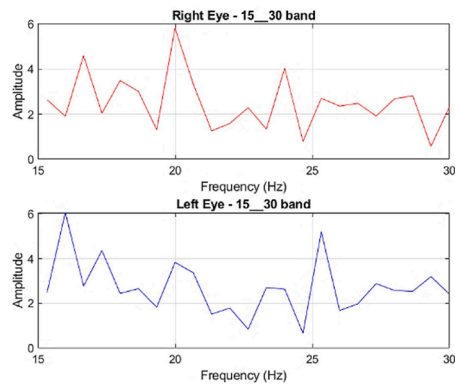


Fig. 8. Frequency plot - (15 to 30 Hz) right and left eye.

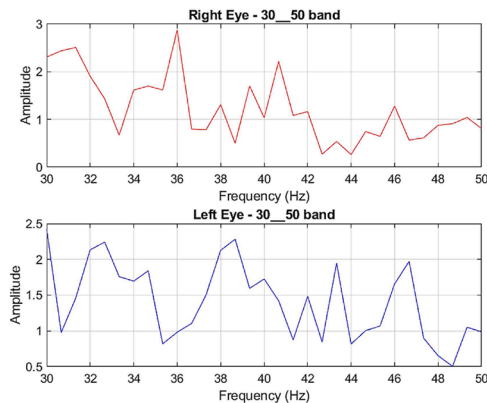


Fig. 9. Frequency plot - (30 to 50 Hz) right and left eye.

extracted, capturing amplitude distribution, signal energy, and temporal variation. These features were selected based on their widespread use in biomedical signal analysis and their relevance for characterizing electrophysiological waveforms [34,35].

Table B.1 (Appendix B) summarizes the extracted features along with their mathematical definitions. While an initial exploratory analysis considered time-domain, frequency-domain, and time-frequency features, statistical features computed from band-limited signals demonstrated the most consistent and interpretable patterns across diagnostic classes. Consequently, the present study focuses on these features to establish a transparent and reproducible classification framework. Analysis involving additional frequency- and time-frequency-domain features is reserved for future work.

Additional analysis involving frequency and time-frequency domain

features will be presented in a subsequent study to expand the scope and diagnostic capabilities.

For each subject, features were extracted separately from the left and right eye signals across the selected frequency bands B_2 (1–5 Hz), B_3 (5–15 Hz), and B_4 (15–30 Hz). The very-low-frequency band B_1 (0–1 Hz) was excluded due to dominance of baseline drift and non-neural fluctuations, while the high-frequency band B_5 (30–50 Hz) was omitted owing to susceptibility to noise and residual oscillatory artifacts. This resulted in 33 features per eye and a combined 66-dimensional feature vector per subject. Feature vectors from all subjects were aggregated into a dataset of size 280×66 , where 280 denotes the total number of PERG recordings. The construction of the feature vectors for individual bands and eyes is formally described in Equations (4)–(10).

Prior to classification, the extracted features were subjected to ANOVA-based post-hoc statistical evaluation, which is described in the subsequent subsection and was not used for feature selection or model training.

2.5. ANOVA-Based feature evaluation for PERG signal discrimination

Analysis of Variance (ANOVA) was employed in this study exclusively as a post-hoc statistical analysis tool to support physiological interpretation of the extracted PERG features. Importantly, ANOVA was not used for feature selection, dimensionality reduction, or model optimization, and it did not influence the training or testing of the machine learning classifiers. All extracted features were retained and provided as input to the classification models.

The primary objective of the ANOVA analysis was to assess the statistical relevance of individual features in differentiating retinal pathology classes and to provide insight into which frequency bands and signal characteristics exhibit stronger group-level separation. One-way ANOVA is well suited for such exploratory analysis in multi-class biomedical datasets, as it compares between-class variance against within-class variance to quantify feature sensitivity across diagnostic categories.

For each feature, the F-statistic was computed as:

$$F = \frac{\text{Between - Group Variance}}{\text{Within - Group Variance}}$$

Higher F-values indicate stronger statistical separation across pathology classes, while lower values suggest overlapping distributions. Corresponding p -values were calculated to assess statistical significance. For visualization purposes, p -values were transformed using the $-\log_{10}(p)$ scale to enhance interpretability of highly significant features.

Statistical significance thresholds were interpreted as follows:

- $p < 0.05$: Statistically significant discrimination
- $p < 0.001$: Very strong discriminatory capability
- $p > 0.05$: No statistically significant difference

Table 3 summarizes within-subject comparisons between left and right eyes. Figs. 10 and 11 illustrate the distribution and relative dominance of statistically significant features across frequency bands and eye regions.

2.5.1. Across-class ANOVA: identifying key PERG features by pathology

Table B.2 (Appendix B) presents the PERG features ranked by one-way ANOVA p -values using pathology class as the grouping variable for comparing pathology classes, focusing on signals in the 1–5 Hz frequency band. From this analysis it can be observed that the features differ significantly across pathology types. The average F -value across all features was 1.61 ± 1.52 , with a mean p -value of 0.49 ± 0.44 and $-\log_{10}(p)$ of 1.64 ± 2.47 . These results highlight that low-frequency features (1–5 Hz) dominate in distinguishing pathology classes.

When selecting the top 20 features with $p < 0.05$, the values improved to $F = 3.59 \pm 1.22$, $p = 0 \pm 0.01$, and $-\log_{10}(p) = 4.79 \pm 2.37$.

Table 3Ranking of PERG features by one-way ANOVA p -values using within-subject comparisons of mean values between left and right eyes.

Rank	Frequency Band (Hz)	Feature	f value	p value	$-\log_{10}(p)$
1	15 to 30	peak2peak	20.4503	7.24×10^{-6}	5.14
2	15 to 30	skewness	19.34578	1.27×10^{-5}	4.90
3	15 to 30	std	18.71658	1.75×10^{-5}	4.76
4	15 to 30	kurtosis	14.7954	0.000131	3.88
5	5 to 15	kurtosis	14.54964	0.000149	3.83
6	5 to 15	skewness	12.92097	0.000349	3.46
7	15 to 30	rms	9.996363	0.001639	2.79
8	15 to 30	freq-variance	9.045048	0.002733	2.56
9	15 to 30	variance	9.045048	0.002733	2.56
10	15 to 30	slope	8.240523	0.004226	2.37
11	15 to 30	mav	5.525829	0.019027	1.72
12	15 to 30	mean	5.525829	0.019027	1.72
13	15 to 30	auc	5.208079	0.022795	1.64
14	5 to 15	auc	4.932458	0.02669	1.57
15	5 to 15	mav	4.790162	0.028967	1.54
16	5 to 15	mean	4.790162	0.028967	1.54
17	5 to 15	rms	4.20554	0.040681	1.39
18	5 to 15	peak2peak	3.777025	0.052379	1.28
19	5 to 15	std	3.542258	0.060257	1.22
20	5 to 15	slope	1.051963	0.305426	0.52
21	1 to 5	slope	0.681835	0.40925	0.39
22	1 to 5	kurtosis	0.244031	0.621472	0.21
23	1 to 5	skewness	0.170409	0.67988	0.17
24	1 to 5	auc	0.161656	0.687765	0.16
25	1 to 5	mav	0.14937	0.699261	0.16
26	1 to 5	mean	0.14937	0.699261	0.16
27	5 to 15	freq-variance	0.120309	0.728809	0.14
28	5 to 15	variance	0.120309	0.728809	0.14
29	1 to 5	freq-variance	0.075404	0.78371	0.11
30	1 to 5	variance	0.075404	0.78371	0.11
31	1 to 5	rms	0.026971	0.869601	0.06
32	1 to 5	peak2peak	0.012073	0.912539	0.04
33	1 to 5	std	0.011294	0.915396	0.04

These features are visualized in Fig. 10, showing their strength in separating pathology classes.

Among these, 14 features with $p < 0.001$ had even stronger separation, with $F = 4.2 \pm 0.9$, $p = 0 \pm 0$, and $-\log_{10}(p) = 5.95 \pm 1.79$. All of these top-performing features were found in the 1–5 Hz range, confirming that low-frequency components are the most informative for classifying retinal pathologies. These features are visualized in Fig. 11.

Figs. A.1 to A.5 (Appendix A) illustrate box plots of the top five PERG features—such as slope, standard deviation (std), peak-to-peak amplitude, and root mean square (rms)—within the 1–5 Hz frequency band, predominantly derived from the right eye, across various retinal pathology classes. Despite the extremely low p -values (as small as 4.9×10^{-9}) indicating strong statistical significance, the box plots exhibit notable distributional overlap among certain classes, particularly Class 6 (Congenital Stationary Night Blindness), Class 10 (Cone-Rod Dystrophy), Class 13 (Retinitis Pigmentosa), and Class 14 (Normal).

This visual overlap suggests several underlying factors:

1. Intra-class variability, where individuals within the same pathology class exhibit diverse retinal response characteristics due to biological heterogeneity or procedural influences.
2. Feature redundancy, indicating that a single feature may not be sufficient to capture the complex and distinctive signal patterns associated with each pathology.

Although the statistical analysis confirms the discriminative potential of these features, their overlapping distributions underscore the limitations of relying on individual features for precise classification. This highlights the necessity of a multi-feature approach, wherein multiple complementary features collectively enhance the accuracy and robustness of classification models.

Moreover, the observed overlap may be influenced by factors such as natural biological variation, minor differences in electrode placement, or variations in disease stage or progression. These findings emphasize the complexity inherent in interpreting PERG signals and reinforce the importance of a comprehensive and strategically selected feature set for clinical and diagnostic applications.

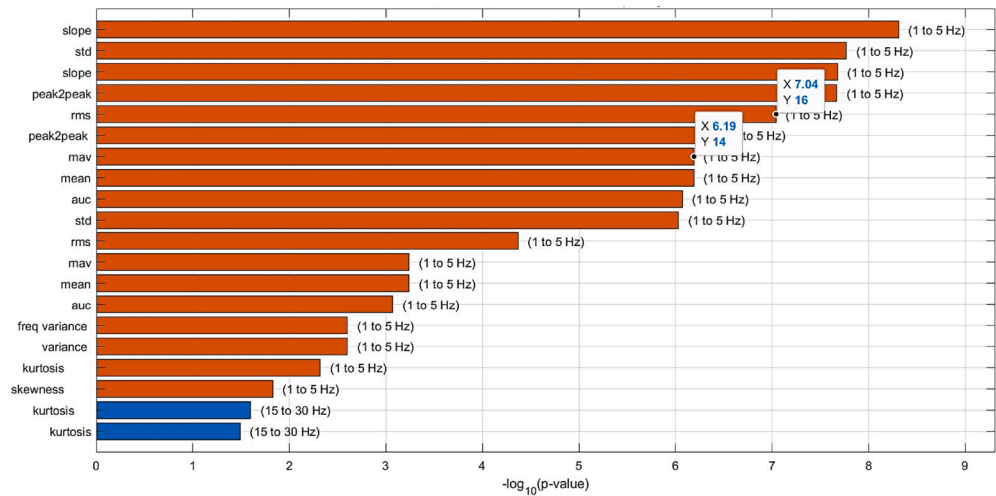


Fig. 10. Significant PERG features by frequency band ($p < 0.05$, $-\log_{10}(p)$ scaled).

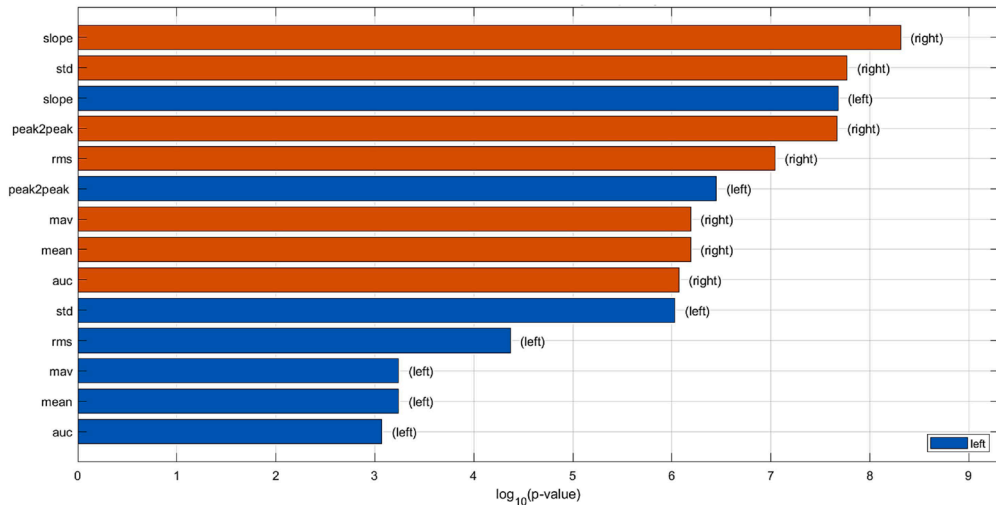


Fig. 11. Significant PERG Features by eye region ($p < 0.001$, $-\log_{10}(p)$ scaled).

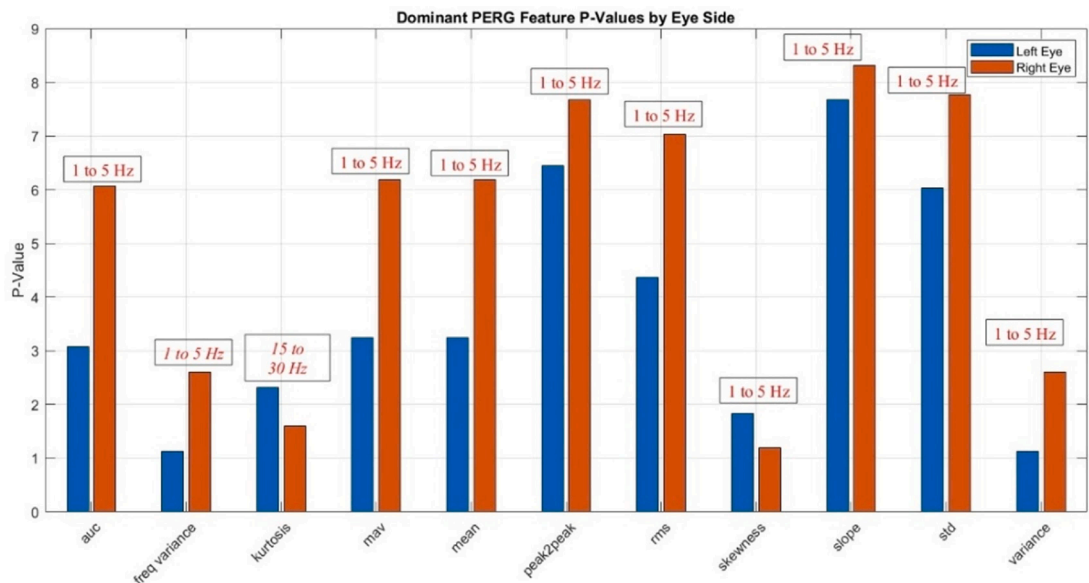


Fig. 12. Dominant PERG features across frequency bands (Features by $-\log_{10}(p)$).

2.5.2. Eye region comparison: left vs. right eye analysis

Table 3 focuses on differences between the left and right eyes for each subject. This within-subject comparison helps identify eye-specific variations in PERG signals.

Across all features, the mean F -value was 5.53 ± 6.29 , $p = 0.31 \pm 0.36$, and $-\log_{10}(p) = 1.58 \pm 1.58$. However, among the 17 features with $p < 0.05$, values improved to $F = 10.12 \pm 5.62$, $p = 0.01 \pm 0.01$, and $-\log_{10}(p) = 2.79 \pm 1.3$.

A subset of 10 highly significant features ($p < 0.001$) showed even greater differentiation between eyes, with $F = 13.71 \pm 4.62$, $p = 0 \pm 0$, and $-\log_{10}(p) = 3.62 \pm 1.05$. These are presented visually in Fig. 12: Dominant PERG Features Across Frequency Bands, with all top-ranked features located in the 15–30 Hz band, showing that higher-frequency signals are more sensitive to left–right differences.

The 10 features with $p < 0.001$ showed mean F -value 13.71 ± 4.62 , p -value 0 ± 0 , and $-\log_{10}(p) = 3.62 \pm 1.05$ —strong lateral differentiation.

Notably, the left eye's mid- to high-frequency bands (5–15 Hz and 15–30 Hz) presented the most pronounced left-right differences, with top features like peak-to-peak, skewness, and std (all in 15–30 Hz) achieving F -values > 14 and $-\log_{10}(p) \geq 3.8$. These findings are summarized in Fig. 12, which maps the dominant PERG features across frequency bands using $-\log_{10}(p)$ scaling. From these analysis of dominant PERG features reveals a distinct pattern regarding the importance of frequency bands and eye regions in two types of evaluations: pathology classification and inter-eye (left vs. right) comparison.

For pathology classification, which evaluates how features vary across different retinal pathology classes, the top-ranked features (see Table B.2 in Appendix B) were consistently located in the 1–5 Hz frequency band. These include:

1. Slope, reflecting signal trend or direction
2. Standard deviation (std), indicating variability
3. Peak-to-peak, showing the signal's amplitude range
4. Root mean square (rms), quantifying energy

These features were mostly derived from the right eye and demonstrated high F -values and extremely low p -values ($p < 0.001$), suggesting strong discriminatory power. This aligns with the physiological role of low-frequency PERG components, which are typically linked to retinal ganglion cell (RGC) response and sustained visual stimuli processing—both of which may be altered in retinal pathologies.

On the other hand, for left–right eye comparison (refer to Table B.2 in Appendix B and Fig. 12), the most discriminative features were concentrated in the 15–30 Hz frequency band from the left eye. These features include:

1. Peak-to-peak
2. Skewness and kurtosis, which describe the distribution shape and signal asymmetry
3. Standard deviation, again reflecting variability

The high-frequency band (15–30 Hz) is generally associated with transient or rapid neural responses, possibly capturing subtle asymmetries in retinal function or electrode placement that manifest differently between eyes. These features were particularly effective in identifying within-subject asymmetries, evidenced by high F -values (up to 20.45) and significant p -values (many < 0.001). These findings suggest that the low-frequency PERG features (1–5 Hz) are more sensitive to changes across pathology classes, supporting their suitability in multi-class classification system. High-frequency features (15–30 Hz) are more sensitive to eye-specific differences, making them valuable for detecting functional asymmetry in clinical evaluations.

This dual-band strategy enhances both diagnostic specificity and clinical relevance, helping practitioners focus on the most informative PERG features depending on the diagnostic goal.

The following section details how the statistically validated features

were post-processed and used to model the classification system, including a description of the classification methodology and evaluation using relevant performance metrics.

3. Machine learning models for retinal pathology classification

Machine learning models were developed using feature sets extracted from PERG signals recorded from both the left and right eyes of each subject. Time-domain features were derived from three predefined frequency bands (1–5 Hz, 5–15 Hz, and 15–30 Hz) independently for each eye, forming the input feature vectors for classification. For each subject, a total of 66 features were extracted and associated with a corresponding diagnostic class label. The resulting feature vectors were organized into a consolidated table structure, with each row representing a unique trial indexed by a generated SubjectID.

The original dataset comprised 15 diagnostic classes ($class_num \in \{1, \dots, 15\}$), each corresponding to a distinct retinal condition. To comprehensively evaluate the classification capability of the extracted PERG features, three classification strategies were implemented:

- (i) binary classification to distinguish normal from pathological cases,
- (ii) multi-class classification using all 15 original labels, and
- (iii) five-class multi-class classification based on clinically motivated regrouping of related conditions.

The five-class grouping consisted of Normal signals (106 records), Retinitis Pigmentosa (47 records), Macular Dystrophy (33 records), Stargardt Disease (16 records), and a consolidated category comprising related retinal and optic nerve disorders with overlapping PERG response characteristics. This regrouping was guided primarily by shared pathophysiological mechanisms and electrophysiological similarities observed in PERG recordings, with class imbalance considered as a secondary factor to support stable classifier training. All evaluations were performed using record-level random splits; subject-wise and cross-dataset validation were not applied and are reserved for future studies. The results corresponding to each classification strategy are presented and discussed in the subsequent sections.

3.1. Data normalization

Prior to training, all feature values were normalized using Min-Max scaling (range normalization) to provide uniformity in feature contribution and accelerate model convergence. Normalization was performed after class filtering but prior to data splitting, using MATLAB's `normalize` function with the 'range' method.

3.2. Classification models

A total of Five different machine learning classifiers were employed to evaluate the discriminative potential of the extracted features. This study evaluates five supervised machine learning classifiers for multi-class classification of PERG signals using a consistent experimental protocol. These classifiers were selected to represent a diverse set of learning paradigms, including multi layer neural networks, ensemble methods, and boosting techniques, allowing for a comprehensive assessment of model robustness and generalizability. A similar approach was adopted in prior work by Narasimhan, G., & Victor, A. (2025) [36], which conducted an empirical analysis of heart disease prediction using various datasets and classification procedures, reinforcing the value of comparative evaluation in clinical signal-based classification tasks. Although deep and hybrid learning approaches are increasingly popular in biomedical signal analysis, this study deliberately focuses on classical machine learning models. The primary reasons are the limited size of the available PERG dataset, the need for signal-level interpretability, and the requirement for clinical transparency. Classical models enable

explicit analysis of feature relevance, frequency-band contribution, and eye-specific effects, which are essential for physiological interpretation and clinical acceptance. In contrast, deep learning models typically require substantially larger datasets and often operate as black-box systems, limiting interpretability in early-stage exploratory studies. In this research the first model, a Multi Layer Neural Network (PatternNet), was configured with one hidden layer containing 15 neurons. The hidden neuron were varied from 10 to 15 to visualize the stability of the system across the classification models and 15 neurons have been finally set. The training was performed using the Levenberg–Marquardt back-propagation algorithm ('trainlm'), with softmax output for multiclass prediction via one-hot encoding. Ensemble learning methods included Random Forest, implemented using MATLAB's 'TreeBagger' with 100 trees and out-of-bag prediction, and Bagged Trees, constructed via 'fitensemble' with the method set to 'Bag'. To address class imbalance, the RUSBoost classifier was employed using 'fitensemble' with 100 decision tree learners and random undersampling of majority classes during boosting. The final model, AdaBoost, utilized the Error-Correcting Output Codes (ECOC) framework with decision trees (maximum split of 20) as base learners and a one-vs-all coding scheme, implemented through MATLAB's 'fitcecoc'.

All classifiers were trained and evaluated across 10 independent trials, with data partitioned using an 80–20 train-test split via MATLAB's 'cvpartition'. Performance was assessed using standard classification metrics: accuracy, precision, recall, specificity, F1-score, training time, and model complexity, where complexity was measured based on trainable parameters, number of learners, or ECOC structure depending on the model. Additionally, interpretability was qualitatively labeled for each classifier. Confusion matrices for each trial were saved in both graphical and CSV formats to enable class-specific error analysis. This structured setup supports a rigorous comparative study of neural and ensemble classifiers, offering insights into their relative performance, generalization ability, and suitability for PERG-based multiclass diagnostic tasks.

4. Results and discussion

The following section presents three sets of classification results derived from this study: binary classification distinguishing normal and pathological PERG responses, multiclass classification across fifteen distinct retinal pathologies, and a clinically consolidated five-class classification model. The results are analyzed to assess classifier behavior under increasing task complexity, followed by physiological and clinical interpretation of the observed feature and frequency-band relevance.

4.1. Binary classification: normal vs. pathological

A binary classification procedure have been implemented to classify between normal and pathological PERG responses. Five different classifiers were used for the evaluation: MLNN-LM (Multi Layer Neural

Network with Levenberg–Marquardt training), Random Forest, Bagged Trees, RUSBoost, and AdaBoost. Table 4 summarizes the performance metrics used in this study, namely accuracy, precision, recall, specificity, F1-score, and training time.

Overall, most classifiers demonstrated strong performance in the binary classification setting, with NN-LM, Random Forest, Bagged Trees, and AdaBoost consistently achieving very high classification accuracy and low variability across trials. Similarly, NN-LM reported $99.82\% \pm 0.56$ accuracy and $99.82\% \pm 0.57$ F1-score, indicating both excellent performance and stability. The stability observed across repeated evaluations indicates that the extracted band-limited PERG features are highly effective in separating normal and pathological responses in this simplified diagnostic scenario.

In contrast, RUSBoost underperformed significantly, achieving only $63.39\% \pm 7.11$ accuracy, with a precision of $31.70\% \pm 3.55$ and an F1-score of $38.69\% \pm 2.84$. The poor performance of RUSBoost may be attributed to its reliance on random undersampling, which can remove informative samples from already limited minority classes, thereby reducing learning capacity and increasing performance variability.

The strong performance observed in the binary classification task can be partly attributed to the clear functional separation between normal and pathological PERG signals. While encouraging, this task represents a relatively less complex diagnostic problem, motivating further evaluation under more challenging multiclass conditions to better assess feature generalization and model robustness.

4.2. Multiclass classification: 15-category retinal pathology model

The fifteen-class classification task represents a significantly more challenging scenario due to severe class imbalance and limited samples in several categories, as shown in Table 5. Performance results for this task are presented in Table 6.

Based on the results, the highest classification accuracy was 89.29% , achieved using the AdaBoost classifier. The associated performance metrics included a precision of 67.52 ± 6.18 , recall of 68.83 ± 6.58 , specificity of 99.23 ± 0.35 , F1-score of 68.14 ± 6.20 , and a training time of 0.13 ± 0.08 s. Across classifiers, performance declined compared to the binary setting, reflecting the increased difficulty of distinguishing among multiple retinal pathologies with overlapping electrophysiological characteristics. Among the evaluated models, AdaBoost achieved the highest overall accuracy and comparatively stronger precision and recall, indicating better adaptability to class imbalance. However, reduced performance in other classifiers highlights the limitations imposed by sparse and uneven class distributions, including the presence of classes with very few samples.

These results underscore the challenges of fine-grained multiclass classification using limited PERG datasets and emphasize the need for either larger datasets or clinically meaningful class consolidation to improve stability and interpretability.

Table 4
Performance comparison of five classifiers for binary retinal pathology classification (normal vs. pathological).

Classifier	Accuracy	Precision	Recall	Specificity	F1	Training Time
NN-LM	99.82 $\pm$ 0.6	99.79 $\pm$ 0.66	99.85 $\pm$ 0.5	99.85 $\pm$ 0.5	99.82 $\pm$ 0.6	1.3 $\pm$ 0.9
Random Forest	99.29 $\pm$ 1.25	99.48 $\pm$ 0.89	98.92 $\pm$ 1.92	98.92 $\pm$ 1.92	99.2 $\pm$ 1.41	0.49 $\pm$ 0.33
Bagged Trees	99.64 $\pm$ 0.75	99.73 $\pm$ 0.57	99.48 $\pm$ 1.11	99.48 $\pm$ 1.11	99.6 $\pm$ 0.84	0.72 $\pm$ 0.34
RUSBoost	63.39 $\pm$ 7.11	31.7 $\pm$ 3.55	50 $\pm$ 0	50 $\pm$ 0	38.69 $\pm$ 2.84	0.08 $\pm$ 0.13
AdaBoost	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	0.04 $\pm$ 0.05

Table 5
Class distribution of retinal pathologies by count and percentage.

Class	Count	Percent (%)	Class	Count	Percent (%)	Class	Count	Percent (%)
1	5	1.79	6	7	2.50	11	14	5.00
2	16	5.71	7	8	2.86	12	33	11.79
3	5	1.79	8	8	2.86	13	47	16.79
4	5	1.79	9	10	3.57	14	106	37.86
5	3	1.07	10	12	4.29	15	1	0.36

Table 6
Performance comparison of five classifiers for 15-class retinal pathology classification.

Trial	Accuracy	Precision	Recall	Specificity	F1	Training Time
NN-LM	51.4 ± 10.6	20.8 ± 9.61	23.9 ± 7.24	96 ± 0.8	21.8 ± 7.77	4 ± 2.5
Random Forest	58.8 ± 5.8	14.3 ± 3.31	18.2 ± 3.31	96.6 ± 0.3	16 ± 3.23	0.4 ± 0.1
Bagged Trees	60 ± 8.08	16.9 ± 3.32	20.1 ± 3.65	96.7 ± 0.6	18.3 ± 3.26	0.6 ± 0.1
RUSBoost	17.5 ± 21.3	16.9 ± 20.7	18.3 ± 19.7	94.1 ± 1.3	16.9 ± 20.6	1 ± 0.4
AdaBoost	89.3 ± 5.12	67.5 ± 6.18	68.8 ± 6.58	99.2 ± 0.4	68.1 ± 6.2	0.1 ± 0.1

4.3. Multiclass classification: five-category retinal pathology model

To improve clinical relevance and address class imbalance, the original fifteen pathology labels were consolidated into five broader diagnostic categories. The resulting class distribution is shown in Table 7, and classifier performance is summarized in Tables 8–13.

Table 8 presents the results of 10 trials using the Multilayer Neural Network with Levenberg–Marquardt (MLNN-LM) algorithm for five-class classification. All metrics—accuracy, precision, recall, specificity, and F1-score—are reported in percentage terms. The model achieved high performance, with perfect scores in Trials 1, 7, and 9. An average accuracy of 96.43 % ± 3.04 indicates strong predictive ability and reasonable stability. However, higher variability in precision (±9.56 %) and recall (±8.79 %), especially in Trial 6, points to challenges with class imbalance. The average F1-score of 93.12 % ± 8.85 reflects a good balance between precision and recall. Training time ranged from 3.43 to 7.58 s, averaging 4.61 ± 1.19 s. Overall, the classifier is effective but sensitive to class distribution, suggesting the need for data balancing in future work.

Table 9 summarizes the performance of the Random Forest (Tree Bagger) model across 10 trials for the five-class classification task. All evaluation metrics are expressed in percentages.

The model achieved a maximum accuracy of 96.43 % and a minimum of 76.79 %, with an average of 89.46 % ± 5.8, indicating moderate classification performance with some variability. Precision and recall showed noticeable fluctuations (65.86 % ± 8.94 and 64.61 % ± 7.41, respectively), highlighting occasional difficulty in consistently identifying positive cases. The F1-score averaged 65.13 % ± 7.77, reflecting modest balance between precision and recall. Specificity remained high and consistent at 97.82 % ± 1.21, confirming reliable identification of negative instances. Training time was efficient and consistent, averaging

Table 7
Retinal pathology class distribution after merging:

Value	Count	Percent (%)
1	78	27.86
2	16	5.71
3	33	11.79
4	47	16.79
5	106	37.86

0.29 ± 0.02 s, with a fixed model complexity. Overall, the classifier performed reasonably well but showed sensitivity to data imbalance and class variation.

The Table 10 presents results from 10 trials using the Bagged Trees (Bootstrap Aggregation) model for five-class classification. Performance metrics are expressed as percentages. The model showed consistent accuracy, ranging from 87.50 % to 94.64 %, with an average of 91.61 % ± 2.67, indicating good stability. Precision and recall averaged 68.72 % ± 9.28 and 67.76 % ± 5.87, respectively, suggesting moderate ability to detect positive instances with some variation across trials. The F1-score averaged 68.16 % ± 7.4, reflecting reasonable balance between precision and recall. Specificity remained high at 98.24 % ± 0.54, confirming consistent recognition of negative cases. Training times were efficient, averaging 0.68 ± 0.06 s, with constant model complexity. Overall, the classifier performed reliably, offering a balanced trade-off between accuracy and computational cost.

Table 11 shows the results of 10 trials using the RUSBoost classifier for five-class classification. Compared to other models, RUSBoost delivered substantially lower performance, with an average accuracy of 36.25 % ± 7.53. Precision and recall were also low, averaging 19.05 % ± 10.12 and 21.16 % ± 5.17, respectively, reflecting poor identification of true positives. The F1-score averaged 19.27 % ± 8.43, indicating weak overall classification quality.

Although specificity remained relatively stable at 85.87 % ± 2.29, this reflects the model’s tendency to favor the majority (negative) classes. Training time averaged 0.37 ± 0.24 s, with model complexity fluctuating (average 70 ± 48.3). These results suggest that RUSBoost struggled with the dataset, likely due to severe class imbalance and insufficient representative samples for minority classes.

Table 12 summarizes the performance of the AdaBoost classifier implemented via the ECOC (Error-Correcting Output Codes) framework. The model showed exceptionally high and consistent results, achieving an average accuracy of 99.11 % ± 1.26 across 10 trials. Precision, recall, and F1-score were all highly stable, averaging 82.92 % ± 0.59, 83.15 % ± 0.59, and 83.03 % ± 0.55, respectively. Specificity was nearly perfect at 99.77 % ± 0.32, indicating excellent identification of negative cases. Training time remained minimal (0.05 ± 0.01 s) and model complexity low (5.7 ± 0.48), highlighting the efficiency of this approach. Overall, AdaBoost-ECOC delivered the best balance of performance, consistency, and computational efficiency among all classifiers tested.

Table 13 presents a comparative analysis of five classification models

Table 8

Performance evaluation of retinal pathology classification using multi layer neural.

Trial	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
1	100	100	100	100	100	4.83
2	94.64	94.95	87	98.64	90.8	3.5
3	94.64	97.15	88.18	98.34	92.45	5.1
4	94.64	96.82	94.44	98.45	95.62	4.42
5	98.21	99.13	98.33	99.41	98.73	4.39
6	94.64	68.18	73.33	98.81	70.66	4.23
7	100	100	100	100	100	4.8
8	96.43	96.85	90.83	99.09	93.74	7.58
9	100	100	100	100	100	3.43
10	91.07	95.61	83.61	97.11	89.21	3.81
Average	96.43 ± 3.04	94.87 ± 9.56	91.57 ± 8.79	98.99 ± 0.92	93.12 ± 8.85	4.61 ± 1.19

Table 9

Performance evaluation of retinal pathology classification using random forest.

Trial	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
1	83.93	52.78	54.71	96.67	53.73	0.31
2	91.07	75.17	68.75	98.22	71.82	0.32
3	89.29	71.04	63.76	97.65	67.21	0.28
4	94.64	80	77.78	98.86	78.87	0.27
5	91.07	71.15	70.56	98.21	70.85	0.3
6	96.43	60.98	66.67	99.32	63.7	0.29
7	87.5	70.93	62.46	97.55	66.42	0.27
8	89.29	58.77	62.5	97.73	60.58	0.27
9	94.64	61.18	66.67	98.86	63.81	0.29
10	76.79	56.61	52.28	95.16	54.36	0.29
Average	89.46 ± 5.8	65.86 ± 8.94	64.61 ± 7.41	97.82 ± 1.21	65.13 ± 7.77	0.29 ± 0.02

Table 10

Performance evaluation of retinal pathology classification using bagged trees (bootstrap aggregation).

Trial	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
1	89.29	58.25	59.93	97.71	59.08	0.65
2	92.86	76.79	72.92	98.42	74.8	0.69
3	91.07	73.28	69.32	98.17	71.24	0.77
4	92.86	79.17	75.93	98.48	77.51	0.75
5	92.86	73.89	73.89	98.59	73.89	0.66
6	94.64	58.33	65	99.01	61.49	0.7
7	87.5	54.4	57.86	97.53	56.07	0.7
8	92.86	78.61	70.14	98.41	74.13	0.69
9	94.64	63.43	66.67	98.67	65.01	0.58
10	87.5	71.03	65.97	97.36	68.41	0.63
Average	91.61 ± 2.67	68.72 ± 9.28	67.76 ± 5.87	98.24 ± 0.54	68.16 ± 7.4	0.68 ± 0.06

evaluated over 10 independent trials using key performance metrics: accuracy, precision, recall, specificity, F1-score, and training time.

1. The AdaBoost (ECOC) model outperformed all other classifiers, achieving the highest average accuracy (99.11 % ± 1.26) and

Table 11

Performance evaluation of retinal pathology classification using RUS Boost.

Trial	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
1	21.43	15.91	13.9	83.62	14.84	0.54
2	37.5	6.25	16.67	83.33	9.09	0.02
3	30.36	5.06	16.67	83.33	7.76	0.02
4	42.86	25	26.74	87.88	25.84	0.5
5	32.14	28.68	25.51	86.51	27	0.62
6	48.21	29.39	26.06	89.17	27.63	0.5
7	42.86	27.64	27.59	88.29	27.61	0.49
8	37.5	6.25	16.67	83.33	9.09	0.02
9	35.71	27.92	23.31	87.08	25.41	0.45
10	33.93	18.44	18.47	86.11	18.46	0.53
Average	36.25 ± 7.53	19.05 ± 10.12	21.16 ± 5.17	85.87 ± 2.29	19.27 ± 8.43	0.37 ± 0.24

Table 12

Performance evaluation of retinal pathology classification using AdaBoost (via ECOC framework).

Trial	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
1	100	83.33	83.33	100	83.33	0.05
2	98.21	82.58	83.33	99.52	82.95	0.04
3	98.21	82.41	83.33	99.57	82.87	0.05
4	100	83.33	83.33	100	83.33	0.05
5	100	83.33	83.33	100	83.33	0.06
6	100	83.33	83.33	100	83.33	0.05
7	96.43	81.67	81.48	99.12	81.57	0.04
8	98.21	82.58	83.33	99.52	82.95	0.03
9	100	83.33	83.33	100	83.33	0.05
10	100	83.33	83.33	100	83.33	0.05
Average	99.11 ± 1.26	82.92 ± 0.59	83.15 ± 0.59	99.77 ± 0.32	83.03 ± 0.55	0.05 ± 0.01

Table 13

Performance comparison of five classifiers for 5-class retinal pathology classification (grouped diagnostic categories).

Classifier	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 (%)	Training Time (seconds)
NN-LM	96.4 ± 3	94.9 ± 9.56	91.6 ± 8.8	99 ± 0.9	93.1 ± 8.9	4.6 ± 1.2
Random Forest	89.5 ± 5.8	65.9 ± 8.94	64.6 ± 7.4	97.8 ± 1.2	65.1 ± 7.8	0.3 ± 0
Bagged Trees	91.6 ± 2.7	68.7 ± 9.28	67.8 ± 5.9	98.2 ± 0.5	68.2 ± 7.4	0.7 ± 0.1
RUSBoost	36.3 ± 7.5	19.1 ± 10.1	21.2 ± 5.2	85.9 ± 2.3	19.3 ± 8.4	0.4 ± 0.2
AdaBoost	99.1 ± 1.3	82.9 ± 0.59	83.2 ± 0.6	99.8 ± 0.3	83 ± 0.6	0.1 ± 0

exceptional consistency across all metrics. Its precision, recall, and F1-score remained tightly clustered around 83 %, with minimal variability, while requiring the least training time (0.05 ± 0.01 s) and low model complexity, making it both highly effective and computationally efficient.

2. The Neural Network with Levenberg–Marquardt (NN-LM) also delivered strong performance, with an average accuracy of $96.43 \% \pm 3.04$ and the highest precision ($94.87 \% \pm 9.56$) among all models. However, its training time was significantly higher (4.61 ± 1.19 s), indicating greater computational cost.

3. Bagged Trees provided a good balance, achieving $91.61 \% \pm 2.67$ accuracy with moderate precision and recall values around 68 %, and reasonable training efficiency. In contrast, Random Forest (Tree Bagger) showed slightly lower accuracy ($89.46 \% \pm 5.80$) and higher variability across precision and recall, reflecting less stable behavior across trials.

4. The RUSBoost classifier, which incorporates random undersampling to address class imbalance, underperformed significantly in all metrics, with accuracy at $36.25 \% \pm 7.53$ and an F1-score of just $19.27 \% \pm 8.43$. While the intention of addressing imbalance was

valid, the aggressive undersampling likely led to information loss, reducing classification quality.

Overall, AdaBoost via ECOC emerged as the most robust and efficient model, followed by NN-LM for accuracy-focused scenarios and Bagged Trees for balance between performance and training time. RUSBoost was not suitable in this context due to its consistently poor results.

The confusion matrices of the AdaBoost and MLNN-LM classifiers corresponding to their lowest classification accuracy trials were analyzed to identify specific patterns of misclassification. These are presented in Fig. 13 and Fig. 14, respectively.

From the AdaBoost confusion matrix, it is evident that the class formed by merging original classes 5–15 exhibited a misclassification rate of 11.1 %, corresponding to 2 incorrectly predicted samples. In the case of the MLNN-LM classifier, Class 2 (Retinitis Pigmentosa) showed a misclassification rate of 33.3 % (1 out of 3 samples), while Class 3 (Macular Dystrophy) had a misclassification rate of 11.1 % (1 out of 9 samples). These results highlight specific weaknesses in class-level prediction, particularly for classes with fewer training samples. Across all classifiers, the five-class configuration resulted in improved and more stable performance compared to the fifteen-class scenario. AdaBoost implemented via the ECOC framework consistently achieved the best balance of accuracy, precision, recall, and computational efficiency. The MLNN-LM classifier also demonstrated strong predictive capability but required higher training time, indicating increased computational cost.

Ensemble-based models such as Bagged Trees and Random Forest offered moderate performance with reasonable stability, while RUSBoost again underperformed, suggesting that aggressive undersampling is not well suited for this dataset. Confusion matrix analysis revealed that misclassifications were primarily concentrated in categories with fewer samples, reinforcing the influence of data imbalance on class-level performance.

Beyond numerical performance, these results motivate a deeper examination of why specific frequency bands and signal features contribute differently across pathology classes and eye regions. The following subsections provide physiological and clinical interpretations of these observations.

4.4. Interpretation of frequency band behavior across pathologies

The observed dominance of low-frequency PERG components (1–5 Hz) in pathology classification can be attributed to their strong association with sustained retinal ganglion cell (RGC) activity and macular processing. Pathologies such as glaucoma, retinitis pigmentosa, and macular dystrophies are known to alter slow, sustained neural responses rather than transient high-frequency activity. Consequently, features

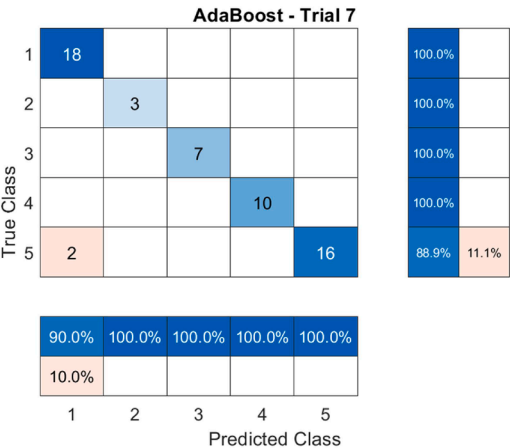


Fig. 13. Confusion matrix of AdaBoost via ECOC with minimum accuracy of 96.43 %.

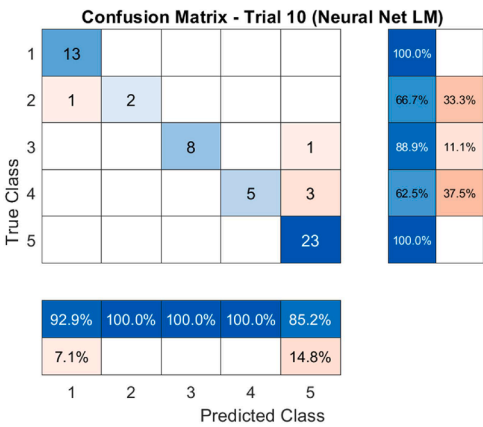


Fig. 14. Confusion matrix of MLNN-LM with minimum accuracy of 91.07 %.

such as slope, standard deviation, peak-to-peak amplitude, and RMS extracted from this band demonstrate greater sensitivity to disease-related functional changes. This explains why low-frequency band features consistently achieved higher statistical significance and stronger discriminative performance across pathology classes.

4.5. Eye-Region and high-frequency feature interpretation

In contrast, features derived from higher frequency bands (15–30 Hz) were more effective in capturing left–right eye differences. High-frequency PERG components are typically associated with transient neural responses and rapid signal fluctuations, which may be more susceptible to subtle asymmetries arising from disease progression, electrode placement, or inter-eye physiological variation. Features such as skewness, kurtosis, and peak-to-peak amplitude in this band were therefore more informative for within-subject comparisons rather than cross-pathology discrimination. This complementary behavior highlights the importance of band-specific feature interpretation depending on the diagnostic objective.

4.6. Clinical and methodological implications

From a clinical perspective, these findings suggest that low-frequency PERG features are more suitable for disease classification tasks, while higher-frequency components may serve as markers of functional asymmetry or disease staging. Methodologically, this reinforces the value of the proposed band-wise analysis framework, which allows frequency-dependent insights that would be obscured in full-band or purely time-domain analysis.

5. Conclusion

This study investigated frequency-band-specific feature behavior in PERG signals for the classification of retinal and optic nerve pathologies. Beyond numerical performance comparisons, the findings provide insight into the physiological relevance of different spectral components. In particular, low-frequency PERG bands demonstrated stronger discriminative capability across pathology classes, which may be attributed to their association with global retinal ganglion cell responses and sustained neural activity. In contrast, higher-frequency features showed greater sensitivity to inter-eye variability, suggesting their potential role in capturing localized or asymmetrical functional changes.

The structured band-wise analysis employed in this work enables systematic comparison of feature relevance across disorders, rather than treating spectral analysis solely as a preprocessing or noise-reduction step. By decomposing the PERG signal into diagnostically meaningful frequency bands and analyzing each band independently, the proposed framework facilitates clearer interpretation of how different spectral

components contribute to disease characterization. This approach complements existing PERG-based diagnostic studies and supports the potential of frequency-domain representations as informative biomarkers when combined with machine learning classifiers.

Despite the promising results, several limitations of the present study should be acknowledged. First, this work represents an exploratory analysis conducted on a limited dataset, which may restrict the generalizability of the reported findings. Subject-wise and cross-dataset validation were not performed, and therefore inter-subject variability and dataset bias could not be fully assessed. Second, the proposed framework relies on predefined frequency bands and handcrafted features, which, while interpretable, may not capture all relevant signal characteristics. Finally, detailed physiological component-level analysis of PERG waveforms (such as N35, P50, and N95) was beyond the scope of the current study.

These limitations highlight the need for further validation using larger and more diverse datasets, as well as the exploration of optimized frequency-band selection and hybrid feature representations.

Ethical statement for computer methods and programs in biomedicine

Hereby, I /Sathees Kumar Nataraj/ consciously assure that for the manuscript / Empirical Analysis of Machine Learning Classifiers for Retinal and Optic Nerve Disorder Detection Using Band-Limited PERG Features/ the following is fulfilled:

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Ethics statement

The dataset used in this study, **PERG-IOBA**, is publicly available on

PhysioNet and was developed by Fernández et al. [4]. The data can be accessed freely at: <https://doi.org/10.13026/d24m-w054>. The authors did not generate or modify the dataset. All preprocessing scripts and analytic code developed for this study are available from the corresponding author upon reasonable request.

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Declaration of generative AI and AI-assisted technologies in the writing process

The author acknowledges the use of AI-assisted tools, specifically ChatGPT by OpenAI, to enhance the linguistic clarity, correct grammatical errors, and improve the overall coherence of the manuscript. All scientific content, analysis, and conclusions presented are entirely the original work of the author. AI assistance was used solely for refining the language and presentation of the text. This usage aligns with the journal's policies on transparency and ethical authorship practices.

CRediT authorship contribution statement

Sathees Kumar Nataraj: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

[Fig. A1,A2,A3,A4,A5](#)

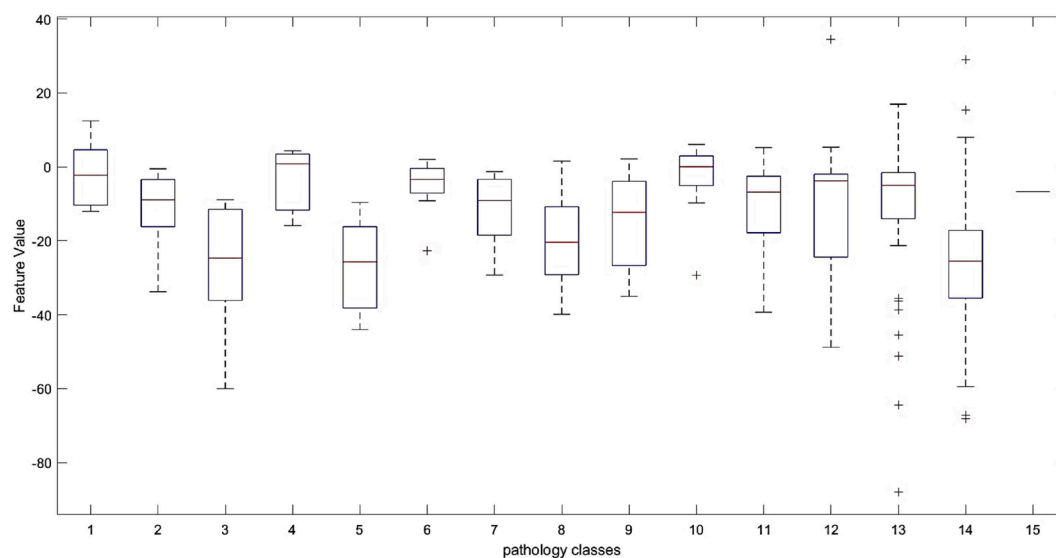


Fig. A.1. Box plot of right eye PERG slope (1-5 Hz) across pathology classes ($p < 4.9 \times 10^{-9}$).

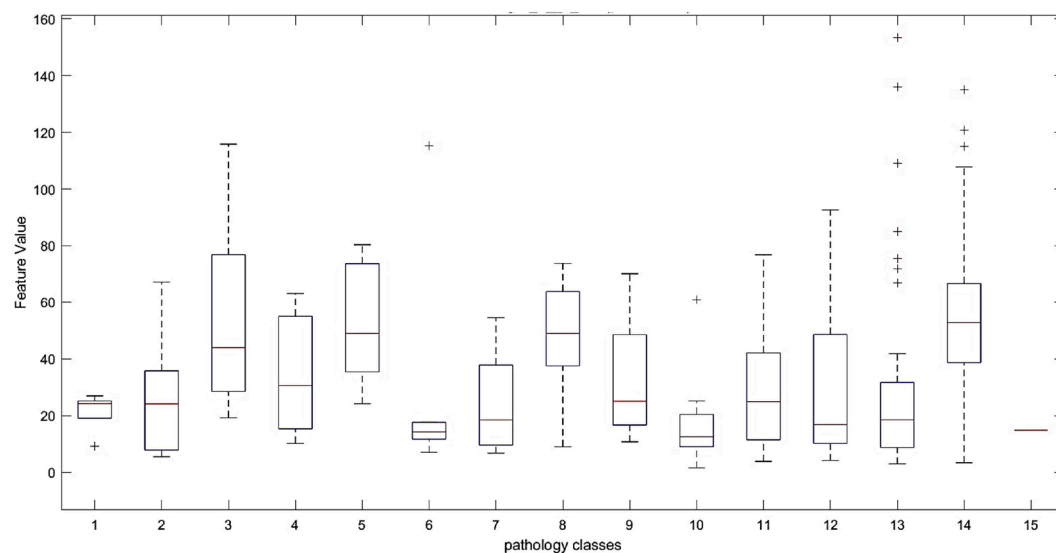


Fig. A.2. Box plot of right eye PERG std (1-5 Hz) across pathology classes ($p < 1.7 \times 10^{-8}$).

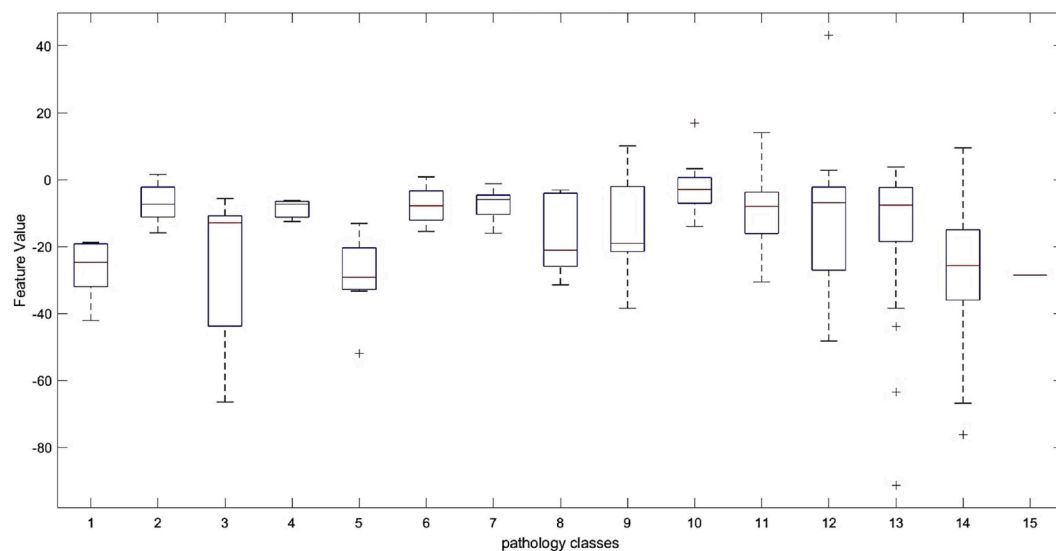


Fig. A.3. Box plot of right eye PERG slope (1-5 Hz) across pathology classes ($p < 2.1 \times 10^{-8}$).

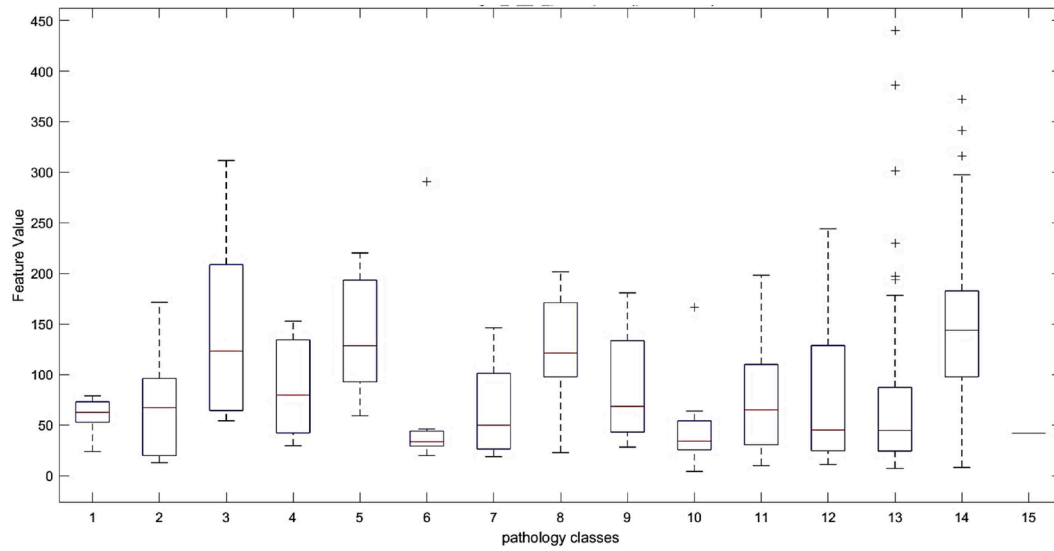


Fig. A.4. Box plot of right eye PERG peak2peak (1–5 Hz) across pathology classes ($p < 2.1 \times 10^{-8}$).

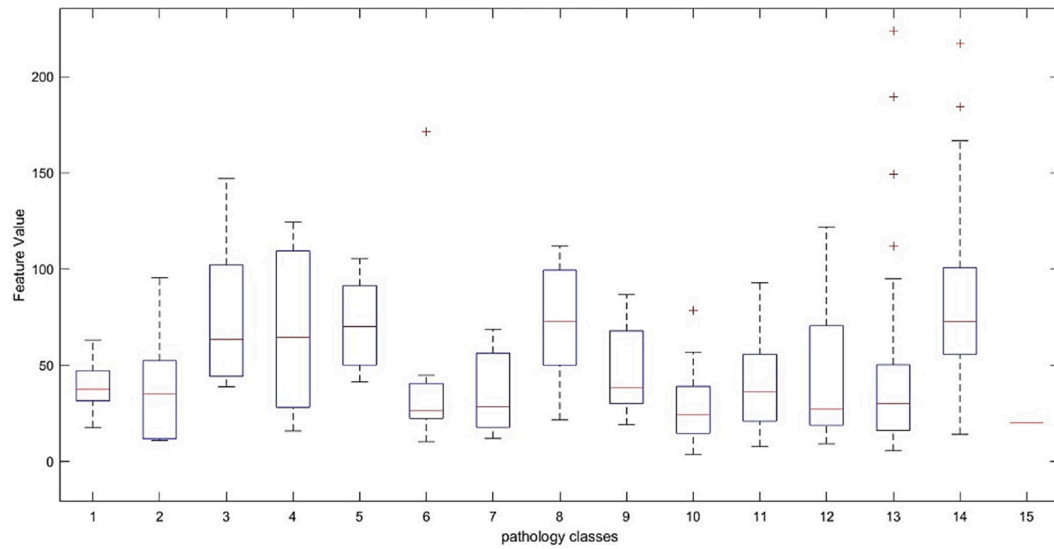


Fig. A.5. Box plot of right eye PERG rms (1–5 Hz) across pathology classes ($p < 9.1 \times 10^{-8}$).

Appendix B

Table B.1

Statistical features extracted from band-limited PERG signals.

Sl. No.	Feature	Mathematical model	Description
1	Mean	$\mu = \frac{1}{N} \sum_{k=1}^N X_i[k]$	Average signal amplitude
2	Variance	$\sigma^2 = \frac{1}{N} \sum_{k=1}^N (X_i[k] - \mu)^2$	Amplitude dispersion
3	Standard deviation	$\sigma = \sqrt{\sigma^2}$	Signal variability
4	Frequency variance	$\mu_f = \frac{\sum_{k=1}^N f_k \cdot x_k}{\sum_{k=1}^N x_k}$ $frequency\ variance = \frac{\sum_{k=1}^N (f_k - \mu_f)^2 \cdot x_k}{\sum_{k=1}^N x_k}$ f_k is the k^{th} frequency bin x_k is the spectral magnitude at f_k μ_f is the spectral centroid	Power-weighted variance of frequency components

(continued on next page)

Table B.1 (continued)

Sl. No.	Feature	Mathematical model	Description
5	Root Mean Square (RMS)	$\sqrt{\frac{1}{N} \sum (X_i[k])^2}$	Signal energy
6	Peak-to-Peak	$\max(X_i) - \min(X_i)$	Dynamic amplitude range
7	Skewness	$skew(X_i)$	Distribution asymmetry
8	Kurtosis	$kurt(X_i)$	Distribution peakedness
9	Slope	$mean(\Delta Xi) = \frac{1}{N-1} \sum_{k=1}^{N-1} \Delta x_k$ $\Delta x_k = x_{k+1} - x_k$	Mean first-order derivative
10	Mean Absolute Value	$\frac{1}{N} \sum_{k=1}^N (X_i[k] - \mu) $	$X_i[k]$
11	Area Under Curve (AUC)	$\int_{f_{low}}^{f_{high}} X(f)df$	Signal magnitude summary

Table B.2
Ranking of PERG features by one-way ANOVA p-values using voice pathology class as the grouping variable.

Rank	Eye region	Frequency Band (Hz)	Feature	f value	p value	Rank	Eye region	Frequency Band (Hz)	Feature	f value	p value	Rank	Eye region	Frequency Band (Hz)	Feature	f value
1	right	1 to 5	slope	5.381	4.88X10 ⁹	23	left	1 to 5	freq variance	1.616	7.47X10 ²	45	left	5 to 15	rms	0.478
2	right	1 to 5	std	5.11	1.71X10 ⁸	24	left	1 to 5	variance	1.616	7.47X10 ²	46	left	15 to 30	auc	0.46
3	left	1 to 5	slope	5.068	2.07X10 ⁸	25	right	15 to 30	slope	1.459	1.26X10 ¹	47	left	15 to 30	freq variance	0.435
4	right	1 to 5	peak2peak	5.061	2.15X10 ⁸	26	right	5 to 15	freq variance	1.369	1.68X10 ¹	48	left	15 to 30	variance	0.435
5	right	1 to 5	rms	4.748	9.12X10 ⁸	27	right	5 to 15	variance	1.369	1.68X10 ¹	49	left	15 to 30	mav	0.43
6	left	1 to 5	peak2peak	4.452	3.59X10 ⁷	28	left	15 to 30	slope	1.368	1.69X10 ¹	50	left	15 to 30	mean	0.43
7	right	1 to 5	mav	4.324	6.50X10 ⁷	29	right	1 to 5	skewness	1.266	2.29X10 ¹	51	right	5 to 15	mav	0.426
8	right	1 to 5	mean	4.324	6.50X10 ⁷	30	right	5 to 15	slope	1.245	2.43X10 ¹	52	right	5 to 15	mean	0.426
9	right	1 to 5	auc	4.263	8.58X10 ⁷	31	right	1 to 5	kurtosis	1.185	2.87X10 ¹	53	right	15 to 30	std	0.392
10	left	1 to 5	std	4.245	9.36X10 ⁷	32	right	5 to 15	peak2peak	1.133	3.29X10 ¹	54	right	5 to 15	auc	0.383
11	left	1 to 5	rms	3.411	4.26X10E5	33	right	5 to 15	std	1.065	3.90X10 ¹	55	left	15 to 30	rms	0.38
12	left	1 to 5	mav	2.826	5.77X10E4	34	left	5 to 15	kurtosis	1.038	4.16X10 ¹	56	left	15 to 30	std	0.374
13	left	1 to 5	mean	2.826	5.77X10E4	35	left	5 to 15	skewness	0.855	6.09X10 ¹	57	left	5 to 15	std	0.367
14	left	1 to 5	auc	2.74	8.42X10E4	36	right	5 to 15	kurtosis	0.816	6.52X10 ¹	58	right	15 to 30	freq variance	0.358
15	right	1 to 5	freq variance	2.484	2.52X10E3	37	left	5 to 15	auc	0.753	7.19X10 ¹	59	right	15 to 30	variance	0.358
16	right	1 to 5	variance	2.484	2.52X10E3	38	right	5 to 15	skewness	0.746	7.26X10 ¹	60	right	15 to 30	auc	0.346
17	left	1 to 5	kurtosis	2.33	4.79X10E3	39	right	5 to 15	rms	0.714	7.60X10 ¹	61	left	5 to 15	peak2peak	0.341
18	left	1 to 5	skewness	2.051	1.48X10 ²	40	left	5 to 15	mav	0.646	8.25X10 ¹	62	right	15 to 30	mav	0.326
19	right	15 to 30	kurtosis	1.912	2.53X10 ²	41	left	5 to 15	mean	0.646	8.25X10 ¹	63	right	15 to 30	mean	0.326
20	left	15 to 30	kurtosis	1.845	3.27X10 ²	42	left	15 to 30	peak2peak	0.505	9.30X10 ¹	64	right	15 to 30	rms	0.323
21	right	15 to 30	skewness	1.665	6.28X10 ²	43	left	5 to 15	freq variance	0.489	9.38X10 ¹	65	right	15 to 30	peak2peak	0.323
22	left	15 to 30	skewness	1.644	6.76X10 ²	44	left	5 to 15	variance	0.489	9.38X10 ¹	66	left	5 to 15	slope	0.302

Average	1.61 ± 1.52	0.49 ± 0.44	1.64 ± 2.47
Average (p < 0.05)	3.59 ± 1.22	0 ± 0.01	4.79 ± 2.37
Average (p < 0.001)	4.2 ± 0.9	0 ± 0	5.95 ± 1.79

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